

Agentic AI Bootcamp

Agentic AI Safety and Guardrails

MAJ Sean Coffey

Army Cyber Institute, U.S. Military Academy

March, 2026

ACI Data & Decision Sciences

Learning Objectives



1

By the end of this module, you will be able to:

- **Implement Multi-Layer Guardrails:** Design input, reasoning, and output filters.
- **Master 2026 Metrics:** Utilize the Agent Value Multiple (AVM) and Task Success Rate (TSR).
- **Align with NIST Standards:** Apply the "Identity and Authorization for AI Agents" framework.
- **Architect Fail-Safes:** Design robust kill-switches and human-in-the-loop triggers.

The 2026 Safety Landscape: Governance at Runtime



2

New Regulatory Realities:

- **NIST AI RMF 2.0:** Shifts focus to "autonomous risk."
- **OWASP Top 10 for Agents:** Highlights "Broken Agentic Authorization."
- **Identity-Centric Safety:** Agents now require verifiable non-human identities (NHI).

The 2026 Challenge

"Deterministic security for non-deterministic systems."

Multi-Layer Guardrail Architecture



3

- ❶ **Input Stage:** Jailbreak detection and PII redaction.
- ❷ **Process Stage (Logic):** Task decomposition checks and "Reasoning Boundaries."
- ❸ **Output Stage:** Groundedness checks (Does it cite sources?) and policy alignment.

Performance Metrics: Beyond Accuracy



4

Metric	Description
Agent Value Multiple (AVM)	Value generated / Total inference + op
Goal Success Rate (GSR)	% of sessions where all sub-goals were
Tool Selection Accuracy	Correctness of tool choice vs. ground t
Reasoning Latency	Time spent in "Thought" vs. "Action"

Success is no longer just a correct answer; it is a reliable process.

The Agent Kill-Switch & HITL



Fail-Safe Design Patterns

- **Recursive Loop Detection:** Automatic termination after N failed tool attempts.
- **Value Thresholds:** Human approval required for actions exceeding \$500.
- **Semantic Divergence:** Halting when the agent's plan deviates from the system prompt.

Human-in-the-Loop (HITL) is not a bottleneck; it is a governance layer.

Summary & Next Steps



- Safety is an **engineering requirement**, not a post-processing filter.
- Use **structured metrics** like AVM to justify agentic spend.
- **Next Block:** We will conclude with "Additional Topics," including Agentic RAG and scaling to multi-agent swarms.